

PRISM-ANT Patent Specification Draft

Engineering claim brief | Not legal advice or a novelty conclusion

Important: This is an engineering invention brief for discussion with patent counsel. It does not establish novelty, freedom to operate, validity, or filing readiness.

1. Proposed technical problem

A shared EV station must respond to arrivals, completions, urgency, and changing battery state while using safe capacity efficiently. A conventional priority score can obscure why power was granted and can improperly trade safety against urgency.

2. Proposed invention

A station controller constructs a per-session model-provenance vector, generates a non-bypassable feasible power envelope, ranks only feasible sessions, allocates station power with capped water-filling, and records the causal explanation. Medical emergency handling changes service priority but retains temperature and electrical safety boundaries.

3. Candidate claim elements

Element	Candidate technical limitation
Signal provenance	Store the source model output, confidence, timestamp, and normalized parameter used in each allocation.
Ordering of control	Apply predictive thermal/health safety mask before policy scoring and allocation.
Forecast-deadline interlock	Use forecast pressure multiplied by individual readiness risk, rather than a station-wide equal priority increment.
Emergency semantics	Bypass battery-life preference only; keep temperature, future trajectory, charger, vehicle, and feeder constraints binding.
Event allocation	Re-run allocation upon arrival, completion, safety state change, or emergency change and reassign released capacity.
Explanation record	Output commanded power, cap source, score terms, and per-model signals for the same decision event.

4. Example independent-claim direction

A computer-implemented method for allocating charging power among electric vehicles at a station, comprising: receiving session telemetry and a service request; generating a predictive signal vector from plural model outputs; determining a maximum feasible power cap for each vehicle from thermal and electrical constraints before assigning an allocation priority; calculating an allocation priority from at least energy need, departure readiness risk, flexibility, forecast-readiness interlock, and policy value; distributing station capacity among feasible vehicles subject to each maximum feasible power cap; repeating the distribution in response to a state event; and storing an explanation record identifying the signals and cap used for each distribution.

5. Dependent-claim directions

Emergency service that bypasses aging preference but preserves thermal masks; a future-temperature trajectory cap; health-confidence gating; a model-version and input-freshness field; a completion event that sets cap to zero; and an allocation log rendering a human-readable reason are potential dependent directions.

6. Prior-art awareness and differentiation hypothesis

Published EV charging patents already describe dynamic current/power allocation using SOC, temperature, time, priority, and session attributes. The candidate differentiator should therefore be evaluated as the **specific ordered combination** of per-model provenance, safety masking before strategy ranking, forecast-readiness interlock, narrow emergency semantics, capped water-filling, and linked explanation record - not merely dynamic allocation or temperature-aware priority.

References for counsel review: US 11,433,772 B2, "Dynamic allocation of power modules for charging electric vehicles"; US 9,469,211 B2, "Managing electrical current allocation between charging equipment"; US 8,643,330 B2, "Method of operating a multiport vehicle charging system". These references demonstrate why a professional prior-art search and claim chart are essential.

7. Evidence to preserve

Keep versioned algorithm source, input/output schemas, model cards, test scenarios, allocation logs, screenshots of explanations, safety tests, and dates of conception/reduction to practice. Do not publish a final claim set until counsel determines publication and filing strategy.

8. Claim-readiness checklist

Define the implementation boundary; quantify technical improvement with tests; identify novel data flow rather than marketing language; build a prior-art claim chart; confirm ownership of training data and code; and obtain a qualified patent attorney or agent review.